

MODEL SELECTION - USING NORMAL PRIORS AND PREDICTIVE SAMPLE RE-USE

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Summary : The predictive sample re-use (PSR) data analysis technique proposed by Geisser and Eddy (1979) is extended to the normal prior case. To obtain this goal, empirical Bayes estimation for the unequal variance case is considered.

1. Introduction

A new, and rather exciting, alternative to classical hypothesis testing and model building, in which significance level and power both vary with sample size, has been proposed by Geisser and Eddy (1979). (See also Geisser (1971), (1975)). They used the method of predictive sample re-use and a Bayesian predictive density derived from a vague prior distribution to determine which of m different models yields the best predictions for future observations. The "quasi-Bayes criterion" is then used to find the most appropriate of the models being considered. This criterion picks the model for which the product of the estimated predictive densities is a maximum for a given set of data.

In many cases however, proper priors will be more appropriate than vague priors. The Stein-estimator (James and Stein (1961)), for example, can be derived from a normal prior density. Other examples where normal priors were used for estimation purposes, are given in Morris (1983), Fay and Herriot (1979), Carter and Rolph (1974) and Hui and Berger (1983). The purpose of this note is therefore to extend the model selection theory of Geisser and Eddy to the normal prior case. In the process of deriving the predictive density, empirical Bayes estimation for the unequal variance case is considered.

In paragraph 5 two real examples are given and in the last paragraph a simulation study is conducted. From the obtained results it is clear, that when appropriate, model selection based on normal priors has good power in detecting the correct model.

2. The Bayes structure in model selection

Consider k samples of the same size N from independent normal distributions with common variance σ^2 and unknown means θ_i , $i=1, \dots, k$.

Thus given θ_i ,

$$(2.1) \quad \bar{X}_i | \theta_i \stackrel{\text{ind.}}{\sim} N(\theta_i, V), \quad i=1, 2, \dots, k$$

where

$$\bar{X}_i = \frac{1}{N} \sum_{j=1}^N X_{ij} \quad \text{and} \quad V = \frac{\sigma^2}{N}.$$

In the Bayesian framework the k means, $\theta_1, \dots, \theta_k$, are generated by a prior distribution. If we assume a normal prior density then

$$(2.2) \quad \theta_i | \mu, A \stackrel{\text{ind.}}{\sim} N(\mu, A), \quad i=1, \dots, k.$$

The model selection problem is to find the model out of a number of proposed ones that best describes the data and is best for predicting future observations. If the best model contains only r distinct estimators for the θ_i 's; $1 \leq r \leq k$, then it is assumed that only r values were generated by the prior distribution and that some of the samples come from the same distribution.

So we redefine the prior distribution as

$$(2.3) \quad \theta_i | \mu, A \stackrel{\text{ind.}}{\sim} N(\mu, A), \quad i=1, \dots, r, \quad 1 \leq r \leq k.$$

This means that when a model with $r < k$ populations is considered, some of the k samples would be from the same population and should be combined to form r samples, but not of the same size anymore. In other words, to find empirical Bayes estimators for the parameters, the unequal variances case must be considered.

3. Empirical Bayes in the unequal variances case

Under a specific model with r populations, assume now that

$$(3.1) \quad \bar{X}_i | \theta_i \sim N(\theta_i, V_i)$$

where $V_i = \frac{\sigma^2}{N_i}$, $i=1, \dots, r$, and N_i is the sample size from the i -th population.

The prior distribution of θ_i is given by

$$(3.2) \quad \theta_i | \mu, A \sim N(\mu, A), \quad i=1, \dots, r,$$

and the posterior distribution by

$$(3.3) \quad \theta_i | \bar{x}_i, \mu, A \sim N(\mu + (1-B_i)(\bar{x}_i - \mu); V_i(1-B_i))$$

where

$$(3.4) \quad B_i = \frac{V_i}{V_i + A}.$$

By assuming that the distribution of μ is a Lebesgue measure on $(-\infty, \infty)$ and independent of A , it follows that θ_i given A and the data will be normally distributed (but not independently anymore) with mean

$$(3.5) \quad E(\theta_i | \bar{x}_1, \dots, \bar{x}_r, A) = m + (1-B_i)(\bar{x}_i - m)$$

and variance

$$\text{Var}(\theta_i | \bar{x}_1, \dots, \bar{x}_r, A) = V_i(1-B_i) + B_i^2 \gamma, \quad i=1, \dots, r,$$

where

$$(3.6) \quad \gamma = \left(\sum_{i=1}^r \frac{B_i}{V_i} \right)^{-1} \quad \text{and} \quad m = \gamma \sum_{i=1}^r \frac{B_i \bar{x}_i}{V_i}.$$

To continue the hierarchical Bayes procedure, a vague prior can be placed on A , but this leads to a complicated form and necessitates numerical integration. An empirical Bayes approach will be used instead. This will be achieved by first deriving an approximate unbiased estimator of A , relative to the marginal distribution of the data, which in turn, will be used to estimate the B_i 's given in (3.4). This will give us closed form approximations for the unconditional posterior means and variances. An unbiased estimate for A , based only on the i -th sample and the grand mean is given by

$$(3.7) \quad \hat{A}_i = \frac{1}{r-1} [r(\bar{x}_i - \bar{\bar{x}})^2 - (r-2)V_i - \bar{V}], \quad i=1, \dots, r,$$

with variance

$$(3.8) \quad \text{Var}(\hat{A}_i) = 2 \left(\frac{r-2}{r-1} V_i + \frac{1}{r-1} \bar{V} + A \right)^2, \quad i=1, \dots, r,$$

where

$$\bar{\bar{x}} = \frac{1}{r} \sum_{i=1}^r \bar{x}_i \quad \text{and} \quad \bar{V} = \frac{1}{r} \sum_{i=1}^r V_i.$$

The derivations of equations (3.7) and (3.8) are given in the appendix. The final estimator for A , based on all the data, is then the weighted mean of all the $\hat{A}_i$'s with weights inversely proportional to their standard deviations (from 3.8), so that

$$(3.9) \quad \hat{A} = \frac{\sum_{i=1}^r W_i \hat{A}_i}{\sum_{i=1}^r W_i}$$

where

$$y^+ = \max(0, y),$$

and

$$(3.10) \quad W_i = \frac{r-1}{(r-2)V_i + \bar{V} + (r-1)A}, \quad i=1, \dots, r.$$

Notice that the weights themselves must be estimated by finding an initial estimate for A in (3.10). An iterative procedure can be used until $\hat{A}$ converges as in Morris (1983), but to avoid iteration let $\hat{A}$ in (3.10) be estimated by the unweighted mean of the $\hat{A}_i$'s in (3.7), say $\hat{\bar{A}}$, so that

$$(3.11) \quad \hat{\bar{A}} = S^2 - \bar{V}$$

and

$$(3.12) \quad \hat{W}_i = \frac{r-1}{(r-2)V_i + \bar{V} + (r-1)\hat{\bar{A}}},$$

where

$$S^2 = \frac{1}{r-1} \sum_{i=1}^k (\bar{x}_i - \bar{\bar{x}})^2.$$

The estimator (3.9) is not the maximum likelihood estimator for A , but is similar to the one given by Morris (1983). Other related estimates appear in Fay and Herriot (1979), Carter and Rolph (1974) and Dempster, Schatzoff and Wermuth (1977)). If V_i is substituted for $\bar{V}$ in equations (3.10) and (3.12), the estimate for A as given by Morris (1983), is obtained. The reason for $\bar{V}$ being included in (3.12) is that the θ_i 's (equation (3.5)) are correlated. The estimate of A and thus of B_i will therefore depend on information from the other populations in the form of $\bar{V}$.

Let

$$(3.13) \quad \hat{B}_i = \frac{r-3}{r-1} \frac{V_i}{V_i + A}, \quad i=1, \dots, r,$$

where the constant $\frac{r-3}{r-1}$ helps to correct for the non-linear dependence of B_i on A (Morris (1983)).

It is assumed that θ_i is approximately normally distributed with mean

$$(3.14) \quad \begin{aligned} E[\theta_i | \bar{x}_1, \dots, \bar{x}_r] &= \hat{m} + (1 - \hat{B}_i)(\bar{x}_i - \hat{m}) \\ &= y_i \end{aligned}$$

and variance

$$(3.15) \quad \begin{aligned} Var[\theta_i | \bar{x}_1, \dots, \bar{x}_r] &= V_i(1 - \hat{B}_i) + \hat{B}_i^2 \hat{\gamma} + v_i[\hat{\gamma} + (\bar{x}_i - \hat{m})^2] \\ &= \sigma_i^{*2}, \text{ say,} \end{aligned}$$

where $\hat{\gamma}$ and $\hat{m}$ are given in (3.6) with $\hat{B}_i$ replacing B_i . The variance of $\hat{B}_i$ is approximated by v_i (see appendix) and is given by

$$(3.16) \quad v_i = \frac{2(r-2)}{(r-3)^2} \frac{\hat{B}_i^4}{V_i^2} \left[\frac{1}{r} \sum_{i=1}^r (V_i - \bar{V})^2 + \frac{r-1}{r-2} (\bar{V} - \hat{A})^2 \right].$$

The additional term on the right-hand side of (3.15), as compared to (3.6), is the additional variation due to the estimation of B_i . As mentioned above, it is not precisely true that $\theta_i | \bar{x}_1, \dots, \bar{x}_r$ has a normal distribution,

but θ_i does have a normal distribution for every fixed B_i and if either B_i is estimated without large variance, or if the added source of variation is taken into consideration, as in (3.15), the normal distribution should hold approximately. An upper bound of 0,0833 can be placed on v_i , $i=1,\dots,r$, since this is the maximum variance a unimodal distribution can have on the interval (0,1).

4. Predictive densities and sample re-use

Consider now a specific model, say M_{rs} , under which r distinct populations, $\pi_1, \dots, \pi_r$, are assumed. The index s is used to distinguish between different models with the same value of r . Then the predictive density function of a future observation from population i , say Z_i , is approximately normal, i.e.

$$(4.1) \quad Z_i | \bar{x}_j, j=1, \dots, r, \pi_i, M_{rs} \sim N(y_i; \sigma_i^2 + \sigma_i^{*2}), \quad i=1, \dots, r,$$

where y_i and σ_i^{*2} are given in (3.14) and (3.15). Equation (4.1) is also the likelihood for a given value of Z_i .

To apply predictive sample re-use, or cross-validation, we omit the j -th observation, x_{ij} , from the sample drawn from population π_i and find the predictive density of x_{ij} given the rest of the data. The likelihood of x_{ij} for all $j=1, \dots, N_i$, $i=1, \dots, r$ under model M_{rs} is then calculated.

The predictive density of X_{ij} is then given by

$$(4.2) \quad X_{ij} | \bar{x}_{i(j)}, \bar{x}_1, \dots, \bar{x}_r, \pi_i, M_{rs} \sim N(y_{i(j)}, \sigma_{i(j)}^2 + \sigma_{i(j)}^{*2})$$

where $\bar{x}_{i(j)}$ denotes the mean of sample i with the j -th observation omitted, and $\bar{\bar{x}}_{(ij)}$ denotes the grand mean with x_{ij} omitted, i.e.

$$(4.3) \quad \bar{x}_{i(j)} = \frac{1}{N_i - 1} (N_i \bar{x}_i - x_{ij})$$

and

$$(4.4) \quad \bar{\bar{x}}_{(ij)} = \frac{1}{r} \left[\sum_{t \neq i}^r \bar{x}_t + \bar{x}_{i(j)} \right].$$

From (3.14) and (3.15) it also follows that

$$(4.5) \quad y_{i(j)} = \hat{m}_{(ij)} + (1 - \hat{B}_{i(j)}) (\bar{x}_{i(j)} - \hat{m}_{(ij)})$$

and

$$(4.6) \quad \sigma_{i(j)}^{*2} = \frac{N_i}{N_i - 1} V_i (1 - \hat{B}_{i(j)}^2) + \hat{B}_{i(j)}^2 \hat{\gamma}_{(ij)} + v_{i(j)} (\hat{\gamma}_{(ij)} + (\bar{x}_{i(j)} - \hat{m}_{(ij)})^2)$$

where

$$(4.7) \quad \hat{m}_{(ij)} = \left[\frac{(N_i - 1) \hat{B}_{i(j)} \bar{x}_{i(j)}}{N_i V_i} + \sum_{t \neq i}^r \frac{\hat{B}_{t(ij)} \bar{x}_t}{V_t} \right] \hat{\gamma}_{(ij)},$$

$$(4.8) \quad \hat{\gamma}_{(ij)} = \left[\frac{(N_i - 1) \hat{B}_{i(j)}}{N_i V_i} + \sum_{t \neq i}^r \frac{\hat{B}_{t(ij)}}{V_t} \right]^{-1},$$

$$(4.9) \quad \hat{B}_{i(j)} = \frac{r-3}{r-1} \frac{N_i V_i}{N_i V_i + (N_i - 1) \hat{A}_{(ij)}},$$

$$(4.10) \quad \hat{B}_{t(ij)} = \frac{r-3}{r-1} \frac{V_t}{V_t + \hat{A}_{(ij)}},$$

and $v_{i(j)}$ follows in the same way from (3.16).

Now $\hat{A}_{(ij)}$ in (4.9) and (4.10) is the combined estimate of A , as defined in (3.9), but without the j -th observation in the i -th sample, so that

$$(4.11) \quad \hat{A}_{(ij)} = \frac{1}{W_{(ij)}} \left[W_{i(j)} \hat{A}_{i(j)}^+ + \sum_{t \neq i}^r W_{t(ij)} \hat{A}_{t(ij)}^+ \right],$$

where $\hat{A}_{i(j)}$ is the estimate of A based on only the i -th sample (without x_{ij}) as given in (3.7), so that

$$(4.12) \quad \hat{A}_{i(j)} = \frac{1}{r-1} \left[r(\bar{x}_{i(j)} - \bar{\bar{x}}_{(ij)})^2 - (r-1) \frac{N_i}{N_i-1} V_i - \bar{V}_{(i)} \right],$$

where $\bar{V}_{(i)} = \frac{1}{r} \left[\sum_{t \neq i}^r V_t + \frac{N_i}{N_i-1} V_i \right],$

and $\hat{A}_{t(ij)}$ is the estimate of A based on the t -th sample, but without x_{ij} , where $t \neq i$.

Thus

$$(4.13) \quad \hat{A}_{t(ij)} = \frac{1}{r-1} \left[r(\bar{x}_t - \bar{\bar{x}}_{(ij)})^2 - (r-1) V_t - \bar{V}_{(i)} \right]$$

where $i=1, \dots, r, \quad j=1, \dots, N_i, \quad t=1, \dots, r; \quad t \neq i.$

In the same way it follows that

$$W_{i(j)} = \frac{r-1}{(r-2) \frac{N_i}{N_i-1} V_i + \bar{V}_{(i)} + (r-1)(S_{(ij)}^2 - \bar{V}_{(i)})^+},$$

$$(4.14) \quad W_{t(ij)} = \frac{r-1}{(r-2)V_t + \bar{V}_{(i)} + (r-1)(S_{(ij)}^2 - \bar{V}_{(i)})^+},$$

$$i=1, \dots, r \\ j=1, \dots, N_i, \quad t=1, \dots, r; \quad t \neq i$$

and

$$(4.15) \quad W_{(ij)} = W_{i(j)} + \sum_{t \neq i}^r W_{t(ij)}.$$

Also

$$(4.16) \quad S_{(ij)}^2 = \frac{1}{r-1} \left[\sum_{t \neq i}^r (\bar{x}_t - \bar{\bar{x}}_{(ij)})^2 + (\bar{x}_{i(j)} - \bar{\bar{x}}_{(ij)})^2 \right].$$

The product of the conditional predictive densities under model M_{rs} is then given by

$$(4.17) \quad L(M_{rs}) = \prod_i^r \prod_j^{N_i} \left[2\pi(\sigma^2 + \sigma_{i(j)}^{*2}) \right]^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2}(\sigma^2 + \sigma_{i(j)}^{*2})^{-1} (x_{ij} - y_{i(j)})^2 \right\}.$$

The quasi-Bayes criterion chooses that model out of all models under consideration with the largest $L(M_{rs})$.

The one exception to the calculation of the predictive densities is in the case where all the means are assumed equal, i.e. for M_{11} : $\theta_1 = \theta_2 = \dots = \theta_k$.

In this case the distribution of θ , given the grand mean $\bar{\bar{x}}$, is independent of A , and the predictive density of a future observation Z is given by

$$Z | \bar{\bar{x}}, M_{11} \sim N(\bar{\bar{x}}, \frac{Nk+1}{k} V)$$

and, omitting x_{ij} ,

$$(4.18) \quad X_{ij} | \bar{x}_{(ij)}, M_{11} \sim N(\bar{x}_{(ij)}, (Nk + 1 + \frac{1}{k(N-1)}) \frac{V}{k}).$$

5. Applications

Example 5.1.

In an example from Neter and Wasserman (1974), p. 456, a breakfast cereal manufacturer uses five agents for handling premium distributions for each of its five products. It was desired to study the timeliness with which the premiums were distributed and at random 20 transactions for each agent were selected. The delay (in days) for handling the transactions was determined for each sample case, and the results were as follows:

Table 5.1.

Agent	<i>X</i>	<i>G</i>	<i>L</i>	<i>F</i>	<i>H</i>
Mean	11,55	14,80	21,95	24,05	31,05

Let us say the purpose of the manufacturer in conducting the investigation is twofold. Firstly, he wants to grade the agents according to the timeliness of their distributions into one of three classes, *A* (good), *B* (average) and *C* (slow), and secondly, the classification must be such that an accurate estimate of the expected delay for an agent in a particular class is obtainable.

We are therefore going to consider out of all possible models only those consisting of three populations. In table 5.1 the agents are ordered according to the sample means and it will only be necessary to consider

them in this order, since any other model with $r=3$ populations and a different ordering will have a smaller $L(M_{rs})$ value. The following six models will therefore be considered.

Table 5.2.

Model	Class			$\ell_n L(M_{gs})$
	A	B	C	
M_{31}	X	G	LFH	-214,1
M_{32}	X	GL	FH	-220,1
M_{33}	XG	L	FH	-196,6
M_{34}	X	GLF	H	-216,2
M_{35}	XG	LF	H	-170,8
M_{36}	XGL	F	H	-227,8

From (4.18) $\ell_n L(M_{gs})$, $s=1, \dots, 6$, were calculated and are given in table 5.2. The quasi-Bayes criterion chooses model M_{35} which classifies agents X and G as grade A, L and F as grade B and H as grade C. The variance was taken as $\sigma^2 = 8,83$, the pooled variance of the five samples. Notice from (3.13) that the general formula for B_i only holds for $r > 3$, so in this example the correction factor $\frac{r-1}{r+1}$, which follows from a first-order Taylor expansion of (3.4), was used. From (3.14) and (3.15) the means and variances of the three categories follows as:

$$y_i = 13,19,$$

$$y_2 = 23,00,$$

$$y_3 = 31,03$$

$$\sigma_1^{*2} = 0,2205,$$

$$\sigma_2^{*2} = 0,2205,$$

$$\sigma_3^{*2} = 0,4407$$

By using the Scott-Knott (1974) criterion the same clustering for three populations were obtained.

Example 5.2.

As a second example to illustrate the method of predictive sample re-use for model selection we turn to the Dyestuff Data of Davies (1967, p. 105), previously used by Box and Tiao (1973) and Geisser (1974 and 1975). The data involve laboratory determinations on five sample units from each of six batches.

Table 5.3.

<u>DYESTUFF DATA</u>						
Batch						
	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>
	195	195	140	145	45	120
	230	150	155	40	40	55
	115	205	90	40	195	50
	235	110	160	120	65	80
	225	160	95	180	145	45
Mean	200	164	128	105	98	70

The variance was taken as $\sigma^2 = 2451,25$.

For clustering of these batches in $r=2$ populations the following results were obtained:

Table 5.4.

Model	I	II	$\ell n(M_{2s})$
M_{21}	<i>A</i>	<i>BCDEF</i>	-136,37
M_{22}	<i>AB</i>	<i>CDEF</i>	-133,52
M_{23}	<i>ABC</i>	<i>DEF</i>	-134,57
M_{24}	<i>ABCD</i>	<i>EF</i>	-137,08
M_{25}	<i>ABCDE</i>	<i>F</i>	-138,58

It is clear that the quasi-Bayes criterion chooses model $M_{22}:AB|CDEF$ from all possible models consisting of only two populations. In the case of the Scott-Knott procedure this is the final grouping. The predictive sample re-use method on the other hand partitions $CDEF$ into two further groups and yields $M_{35}:AB|CDE|F$ ($\ln L(M_{35}) = -133,08$) as the best model among all possible models. (In fact, there are another five partitions better than M_{22} .)

Although both procedures are cluster analysis methods which partition the groups (batches) into discrete non-overlapping clusters, they will often select different models as being the best. The reason for this is that the quasi-Bayes criterion picks the model for which the product of the estimated predictive densities is a maximum while the Scott-Knott procedure is a testing procedure, starting with the assumption of no difference, and will only partition a group if the difference is considered significant. The Scott-Knott procedure is therefore more conservative than the quasi-Bayes criterion as can be seen from the example. This means that the latter procedure will tend to select models with more clusters than the former. It is therefore not surprising that the quasi-Bayes procedure selects M_{35} as the best model since by using the normal test (σ^2 is assumed to be known) it is found that the p-value of 0,057 is nearly significant when testing the equality of the means of clusters CDE and F .

6. Simulation study

In a simulation study with $k=10$ the following ten possible models were considered:

$$M_1: \theta_1 = \dots = \theta_{10}$$

$$M_2: (\theta_1 = \dots = \theta_4) \neq (\theta_5 = \dots = \theta_{10})$$

$$M_3: (\theta_1 = \dots = \theta_4) \neq (\theta_5 = \theta_6) \neq (\theta_7 = \dots = \theta_{10})$$

$$M_4: (\theta_1 = \theta_2) \neq (\theta_3 = \theta_4) \neq (\theta_5 = \theta_6) \neq (\theta_7 = \dots = \theta_{10})$$

$$M_5: (\theta_1 = \theta_2) \neq (\theta_3 = \theta_4) \neq (\theta_5 = \theta_6) \neq (\theta_7 = \theta_8) \neq (\theta_9 = \theta_{10})$$

$$M_6: (\theta_1 = \theta_2) \neq (\theta_3 = \theta_4) \neq (\theta_5 = \theta_6) \neq (\theta_7 = \theta_8) \neq \theta_9 \neq \theta_{10}$$

$$M_7: (\theta_1 = \theta_2) \neq (\theta_3 = \theta_4) \neq (\theta_5 = \theta_6) \neq \theta_7 \neq \theta_8 \neq \theta_9 \neq \theta_{10}$$

$$M_8: (\theta_1 = \theta_2) \neq (\theta_3 = \theta_4) \neq \theta_5 \neq \theta_6 \neq \theta_7 \neq \theta_8 \neq \theta_9 \neq \theta_{10}$$

$$M_9: (\theta_1 = \theta_2) \neq \theta_3 \neq \theta_4 \neq \dots \neq \theta_{10}$$

$$M_{10}: \theta_1 \neq \theta_2 \neq \dots \neq \theta_{10}.$$

The means were generated by each of the 10 models in turn according to equation (3.2) with $\mu=0$ and $A=1$. Ten samples of size 10 were then drawn from a normal population with mean $\theta_i (i=1, \dots, r)$ and variance $\sigma^2=1$. For example, when M_3 was taken as the true model, three means were generated by $N(0,1)$, and then 10 samples, the first four having the same generated population mean, the next two the second generated mean, while the last four samples are drawn from a population with the third mean. $L(M_r)$ is then calculated for each of the ten possible models and the one for which $L(M_r)$ is a maximum chosen as the best model for the data. This was repeated 100 times for each of the 10 "true" models. The number of times each model was chosen as the best one is given in table 6.1.

Table 6.1

True model	Chosen model									
	M_1	M_2	M_3	M_4	M_5	M_6	M_7	M_8	M_9	M_{10}
M_1	<u>82</u>	5	3	1	1	2	0	1	4	1
M_2	8	<u>69</u>	4	4	1	4	4	2	4	0
M_3	6	18	<u>52</u>	7	4	4	4	3	0	2
M_4	1	4	19	<u>53</u>	5	8	7	0	2	1
M_5	2	3	3	10	<u>60</u>	7	7	5	1	2
M_6	0	0	4	6	17	<u>56</u>	5	4	5	3
M_7	0	1	1	4	10	18	<u>49</u>	8	6	3
M_8	1	0	0	2	3	5	24	<u>46</u>	11	8
M_9	0	0	0	0	2	4	8	11	<u>64</u>	11
M_{10}	0	0	0	0	3	3	2	8	26	<u>58</u>

The discriminatory power of the procedure seems quite good and it selects the correct model on average 59% of the time from the ten models, some of which are very close to each other.

In this experiment it does not seem that there is a significant change in the power of the procedure as the true number of populations changes, except perhaps when there is only one population. We have taken σ^2 equal to A , but we expect the power of the procedure to increase if A gets larger relative to σ^2 .

Appendix

Equations (3.7) - (3.8)

From (3.1) and (3.2) it follows that

$$(A.1) \quad \bar{X}_i | \mu, A \sim N(\mu, A + V_i), \quad i=1, \dots, r$$

and

$$(A.2) \quad \bar{X} | \mu, A \sim N\left(\mu, \frac{1}{r}(A + \bar{V})\right)$$

where

$$\bar{X} = \frac{1}{r} \sum_{i=1}^r \bar{X}_i \quad \text{and} \quad \bar{V} = \frac{1}{r} \sum_{i=1}^r V_i.$$

The covariance between $\bar{X}_i$ and $\bar{X}$ given A is

$$(A.3) \quad \text{Cov}(\bar{X}_i, \bar{X} | A) = \frac{1}{r}(V_i + A)$$

so that

$$(A.4) \quad \bar{X}_i - \bar{X} | A \sim N\left[0, \frac{1}{r}[(r-1)A + (r-2)V_i + \bar{V}]\right], \quad i=1, \dots, r.$$

This means that $(\bar{x}_i - \bar{x})^2$ is an unbiased estimator for $\frac{1}{r}[(r-1)A + (r-2)V_i + \bar{V}]$, from which (3.7) follows.

Equation (3.8) follows by noticing that

$$(A.5) \quad \text{Var}(\hat{A}_i) = \frac{r^2}{(r-1)^2} \text{Var}(\bar{X}_i - \bar{X})^2$$

and

$$(A.6) \quad (\bar{X}_i - \bar{X})^2 \sim \frac{1}{r}[(r-1)A + (r-2)V_i + \bar{V}] \chi_1^2.$$

Equation (3.16)

Since

$$(A.7) \quad \hat{B}_i = \frac{r-3}{r-1} \frac{V_i}{V_i + A},$$

a first-order Taylor approximation to the variance of $\hat{B}_i$ is given by

$$(A.8) \quad \text{Var}(\hat{B}_i) \approx \text{Var}(\hat{A}) \left\{ \frac{\partial}{\partial A} \hat{B}_i \Big|_{\hat{A}} \right\}^2 = \left(\frac{r-1}{r-3} \right)^2 \frac{\hat{B}_i^4}{V_i^2} \text{Var}(\hat{A}).$$

Let $\hat{A}$ be approximated by the unweighted mean of the $\hat{A}_i$'s, that is,

$$(A.9) \quad \hat{A} = \frac{1}{r} \sum_{i=1}^r \hat{A}_i = S^2 - \bar{V},$$

with

$$(A.10) \quad \text{Var}(\hat{A}) = \frac{1}{r^2} \left[\sum_{i=1}^r \text{Var}(\hat{A}_i) + \sum_{i \neq j} \text{Cov}(\hat{A}_i, \hat{A}_j) \right].$$

From (3.8) it follows that

$$(A.11) \quad \begin{aligned} \sum_{i=1}^r \text{Var}(\hat{A}_i) &= \frac{2}{(r-1)^2} \sum_{i=1}^r [(r-2)(V_i + A) + (\bar{V} + A)]^2 \\ &= \frac{2}{(r-1)^2} \left[(r-2)^2 \sum_{i=1}^r (V_i + A)^2 + r(2r-3)(\bar{V} + A)^2 \right]. \end{aligned}$$

Also

$$(A.12) \quad \begin{aligned} \text{Cov}(\hat{A}_i, \hat{A}_j) &= \frac{1}{(r-1)^2} E\{[r(\bar{X}_i - \bar{X})^2 - (r-2)V_i - \bar{V}] \\ &\quad \cdot [r(\bar{X}_j - \bar{X})^2 - (r-2)V_j - \bar{V}]\} - A^2 \\ &= \frac{1}{(r-1)^2} \{r^2 E(Y_i^2 Y_j^2) - r[(r-2)V_i + \bar{V}]E(Y_j^2) \\ &\quad - r[(r-2)V_j + \bar{V}]E(Y_i^2) + [(r-2)V_i + \bar{V}][(r-2)V_j + \bar{V}]\} - A^2 \end{aligned}$$

where

$$Y_i = \bar{X}_i - \bar{X}.$$

From (A.4) follows that

$$(A.13) \quad \text{Cov}(Y_i, Y_j) = \frac{1}{r}(\bar{V} - V_i - V_j - A),$$

and it can be shown that

$$\begin{aligned}
 (A.14) \quad E(Y_i^2 Y_j^2) &= \text{Var}(Y_i) \text{Var}(Y_j) + 2(\text{Cov}(Y_i, Y_j))^2 \\
 &= \frac{1}{r^2} \{ [A + (r-2)V_i + \bar{V}] [A + (r-2)V_j + \bar{V}] \\
 &\quad + 2(\bar{V} - V_i - V_j - A)^2 \}.
 \end{aligned}$$

Substituting into (A.12) yields

$$(A.15) \quad \sum_{i \neq j} \text{Cov}(\hat{A}_i, \hat{A}_j) = \frac{2}{(r-1)^2} \sum_{i \neq j}^r (\bar{V} - V_i - V_j - A)^2$$

$$\begin{aligned}
 (A.16) \quad &= \frac{2}{(r-1)^2} \left[2(r-2) \sum_{i=1}^r V_i^2 + 2r^2 \bar{V}^2 \right. \\
 &\quad \left. - r(r-1)(\bar{V}-A)(3\bar{V}+A) \right].
 \end{aligned}$$

Substitution of (A.11) and (A.16) into (A.10) and simplifying gives

$$(A.17) \quad \text{Var}(\hat{A}) = \frac{2(r-2)}{(r-1)^2} \left[\frac{1}{r} \sum_{i=1}^r (V_i - \bar{V})^2 + \frac{r-1}{r-2} (\bar{V} + A)^2 \right]$$

which in turn will yield (3.16) when substituted into (A.8).

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